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Research paper

User behaviour models to forecast electricity consumption of residential customers based on smart metering data

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ABSTRACT

This paper presents a novel approach to forecast day-ahead electricity consumption for residential households where highly irregular human behaviour plays a significant role. The methodology requires data from fiscal smart meters, which makes it applicable to real scenarios where personal data gathering is not feasible. These data are rarely complete; therefore, a robust combination of machine-learning techniques is used to handle missing data and outliers. The novelty of this method relies on identifying and predicting user electricity consumption behaviour as a procedure to improve the forecasting of the overall electricity consumption of each individual customer. The methodology uses Gaussian mixture clustering to identify behaviour clusters and an eXtreme Gradient Boosting classification (XGBoost) model to predict the day-ahead behaviour pattern. This predicted user behaviour cluster is fed into an Artificial Neural Network (ANN) to enable an improved capturing of the highly unpredictable user conduct for the forecast of electricity consumption. A novel metric, namely the *Euclidean Distance-based Accuracy (EDA)*, is finally proposed to enable a more thorough evaluation of time series classification algorithms. The whole development is tested over 500 residential users placed in a southeastern region of Spain. The results showed that, when the novel approach was used, the MAPE_d and NRMSE_d were reduced by 7% and 9% respectively, increasing to a 20% and 17% respective reduction for the best cases according to *EDA*. This methodology sets the basis for massive user-centred analyses, very profitable to any electricity company.

1. Introduction

Traditionally, forecasting of electricity consumption in the residential sector has focused on predicting the hourly aggregated electricity consumption of hundreds or thousands of households. The forecasts have been used as a fundamental tool for properly managing the existing centralised power system because having the ability to foresee periods of high demand allows switching on additional power plants with sufficient time in advance to avoid massive electrical shortages (Hong et al., 2014). These aggregated forecasting methods have been also extensively used to support the electricity demand purchase in wholesale electricity markets.

Over recent decades, the awareness of the need to make significant reductions in greenhouse gas emissions has grown, and a massive transition towards distributed Renewable Energy Sources (RES) is taking place. In this new distributed power system paradigm, large-power plants are being replaced by many smaller ones. Residential customers are leaving the passive role of "only paying" and started to have a central role by contributing as generators and smartly regulating the periods they consume. Therefore, to guarantee the proper synchronisation of this more complex electricity network, comprised of these new highly fluctuating generators, and to optimise the energy efficiency of the whole system, there is a need to accurately forecast the residential electricity consumption at the individual or disaggregated level. This requires more precise forecasting models able to predict both the RES generation and the electricity consumption at individual or community-based levels (Aman et al., 2015).

When predicting electricity consumption of the residential sector, customers' behaviour is an important influencing factor. This is because individual electric consumption depends heavily

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on the process of people's decision making, which has an inherent stochastic nature (Guo et al., 2021). In forecasting electricity consumption at the aggregated level, the variability of each customer is balanced, and the user behaviour effect is attenuated. However, when developing individual or disaggregated models, the behaviour component of each customer gains importance, leading to more significant variability in the energy usage pattern of the customers. Consequently, at the aggregated level, the attenuation of the user behaviour effect entails significantly smaller prediction errors than at the individual level.

Another challenge when dealing with individual electricity consumption forecasting is related to data gathering and data access. Many previous research works used sub-metered data from appliance consumption or outcomes from customers' surveys to forecast individual electricity consumption. However, this kind of data is difficult to obtain when working in real business scenarios with a large number of residential households (Jiang et al., 2020). For example, gathering daily usage schedules filled in by users is completely infeasible when the number of customers is significant. Moreover, obtaining access to this type of personal data is hardly possible in practice due to data privacy concerns. When data comes from domestic appliances or home energy systems, although their market penetration is rapidly increasing, deploying and installing hardware components is still a complex task when conducting a massive implementation. These are the main reasons why our research is oriented towards using data from fiscal smart meters, enhanced with local weather data, which can be obtained through bilateral agreements with electricity trader companies and weather-forecast web services.

Analysing data coming from fiscal smart meters requires robust methods to handle missing data and outliers since they are frequently present in the data sets. In previous research works, this issue was not addressed in detail since most of them did not deal with real data coming from the electricity system. One of the novelties of this paper is a contribution to fill this gap by developing an innovative combination of clustering techniques, classification procedures and machine-learning-based forecasting models, which are capable of achieving a smooth management of these data inconsistencies.

The newly developed approach combines a behaviour model prediction with an electricity consumption forecasting model for individual users. The objective is to forecast hourly, one-day ahead electricity consumption for a large number of residential households (without aggregation), taking into account the significant influence of the stochastic part of user behaviour. It starts with Gaussian mixture clustering to identify behaviour clusters, followed by the implementation of an XGBoost classification model to predict the day-ahead behaviour cluster. This predicted user behaviour pattern becomes the main input for an individual electricity consumption prediction model, based on Artificial Neural Network (ANN) techniques, and enables a more precise capturing of the stochastic component of the behaviour.

The use of a behaviour model, based on time series classification methods, tends to make the interpretation of its accuracy a difficult task. The usual binary metrics for time series accuracy assessment do not provide full understanding when dealing with daily consumption patterns, which is the basic element of classification in our case. Therefore, a new metric, namely the *Euclidean Distance-based Accuracy (EDA)*, is proposed in this research to enable a more thorough evaluation and quantification of time series classification algorithms. This constitutes and important contribution of the paper.

The combined approach, supported by the new developed metric aims at establishing the basis for massive user-centred analyses in the electricity sector. These kind of enriched data-driven methods are expected to be very profitable for electricity network operators, demand response aggregators, or electricity-trading companies in their end-user-based service portfolios.

2. Literature review

Various probabilistic data-driven models have been developed to deal with the energy behaviour of household appliances. Zhao et al. (2014) used office appliances consumption data from an office building to obtain insights into occupant behaviour. Sancho-Tomás et al. (2017) proposed a stochastic modelling of the energy use of small appliances and categorised them in four groups: audio–visual, computing, kitchen and other appliances. Zhang and Guo (2020) presented an ensemble method to forecast the hourly electricity consumption using data from manually-operated devices (such as dishwashers, heat pumps, televisions). In Lan et al. (2014) thousands of electricity consumption profiles were generated from combinations of activation moments from different domestic appliances. Ferrandez-Pastor et al. (2017) presented the design of embedded hardware components able to disaggregate household power consumption. Chou and Tran (2018) used data collected from equipment installed in an experimental building to evaluate the effectiveness of different machine learning techniques. And Haq et al. (2020) used electrical appliance consumption data (fridge, oven, hair dryer, lighting, air conditioner, laptop, water heater, television, iron, cloth dryer) of 10 household to forecast energy consumption.

Other studies used the outcomes of occupants' surveys to strengthen their models. Aerts et al. (2014) developed a probabilistic model to identify typical occupancy patterns based on a time-of-use survey. Liisberg et al. (2016) used Hidden Markov Models combined with surveys to characterise occupant behaviour in a residential building block. Palacios-Garcia et al. (2018) presented a bottom-up stochastic model for electricity demand of heating and cooling appliances. They stated that the occupancy profile is the cornerstone of the model, emphasising that energy consumption in the residential sector is strongly related to people's activity. They fed their model with information gathered in a diary, which had to be completed by people with a frequency of 10 min and had to include the daily activities performed.

All these mentioned authors used data from sub-metering systems able to measure the electricity consumption of domestic appliances in residential households. Some of them also used the outcomes from customers' surveys to improve their prediction models. However, these kind of data are difficult to obtain when working under real scenarios where data from sub-metering systems are unavailable. In this case, fiscal smart meters become a valuable data source; here there is only a single measure of the overall household electricity consumption. Many researchers focused their activity on obtaining electricity consumption patterns and perform customers' segmentation by using these data in clustering algorithms (Al-Otaibi et al., 2016; Al Khafaf et al., 2021; López et al., 2011). Extracting typical consumption patterns helped electricity Distribution System Operators (DSOs) or electricity retailers to develop innovative tariff structures aiming at efficient consumption profiling, as stated by Faria et al. (2016), Chicco et al. (2004), Melzi et al. (2017). In Aghabozorgi et al. (2015), a comprehensive overview of available time series clustering methods and their applications was performed.

Beyond energy behaviour characterisation, day-ahead electricity consumption forecasting of residential users has also been intensively analysed in recent years. Detailed reviews of (Yildiz et al., 2017) and van der Meer et al. (2018) showed that when data comes exclusively from fiscal smart meters, probabilistic data-driven models are the most common approach and that ANNs have become predominant among these techniques. Chou and Tran (2018) did a comprehensive study of the use of different data-driven consumption forecasting models based on energy consumption time series of residential households, concluding again that the best performing data-driven model were the ANNs. The reason for the good performance of ANNs, when being used to predict electricity consumption, is their ability to capture non-linear relationships between input data and target parameters (Mena et al., 2014; Li et al., 2015).

Nevertheless, all the indicated data-driven models, including the ANNs, have some limitations in capturing the erratic, but highly periodic, energy habits of residential customers when the only available data are the electricity consumption, the calendar information and the external weather data. These data sets contain information of intra day user performance, climate dependencies and also weekly, daily and hourly dependencies. However, there is no way to directly infer the user behaviour which is not related to these seasonal dependencies. If a customer decides to turn on the washing machine twice a week when they get home, an ANN or regression based model cannot infer this periodic behaviour with these data inputs. It is necessary to include, at least one new exogenous variable which may have a direct or indirect link with this user behaviour. In previous research work (Mor et al., 2021; Grillone et al., 2021; Yildiz et al., 2018), it was found that a good procedure to define this new input is to find the electricity consumption daily profile that is most likely to occur in the 24 h forecasting horizon, by using a combination of clustering techniques, daily feature extraction and machine learning methods. This is the procedure followed in this research paper and constitutes a significant improvement in relation to previously mentioned electricity consumption forecasting methodologies. This strategy is built on the foundations that human behaviour repeatedly obeys certain daily routines, which should be evidenced in similarities in their electrical consumption curves (Hsiao, 2015).

3. Main contributions

The main contribution of this research work relies on developing and testing a combination of behaviour and electricity consumption modelling to enrich the residential electricity consumption forecasting and its accuracy assessment within large-scale data-based computing procedures based on smart metering data.

The research develops a day-ahead forecasting workflow that inserts information captured by a daily behaviour model into an hourly-based electricity consumption model. The methodology starts by clustering the electricity consumption days, identifying a set of characteristic electricity consumption patterns that will determine the typical behaviours for each user. Then, the XGBoost framework is used to forecast which one of the set of characteristic behaviour patterns will each user perform the following day. The XGBoost uses, as inputs, extracted features of the present and previous days. Both the extracted features of the present day and the predicted day-ahead pattern are finally included as inputs into the hourly electricity consumption forecasting model (ANN).

This novel approach has proven to improve the accuracy of ANN models to predict electricity consumption of residential customers at hourly data granularity, overcoming the barriers imposed by the highly erratic human behaviour. Although ANN techniques have been proven to achieve very good results, they are not capable of capturing user behaviour without including it as an explicit variable. The novel strategy provides an alternative approach to precisely determine the electricity load of a large amount of residential users at individual level. An accurate comparison has been carried out, in which the most widely used hourly consumption forecasting model (ANN) is evaluated with and without including the behaviour model. The results show a considerable improvement introduced when using the novel behaviour model.

The whole methodology was successfully tested over an extensive set of households in a southeastern region of Spain. Although several preceding studies present forecasting models, few of them test their procedures under real massive scenarios. For this case, despite the significant problem generated by the great amount of missing data and outliers (since smart meter readings are rarely complete), the presented methodology proved to work correctly. It is important to remark that these good results were obtained even when 27.5% of the testing data set were incomplete days for the presented case study. This percentage is representative of a real data corpus; which guarantees that the methodology can be used in real cases. Moreover, the development was implemented with a focus on avoiding costly computational tasks, making it work within a reasonable processing time, even for large amounts of users. Thus, supporting the fact that it is very suitable for real business scenarios.

In addition, the behaviour model allows for a better understanding of each customer, because an importance feature analysis can be done to gain knowledge of which are the most significant features that determine each user's daily behaviour. This could be a very useful tool for electricity trader companies.

Finally, analysing the literature, a knowledge gap was found regarding metrics to evaluate the prediction accuracy of time series forecasting models when using classification techniques. Therefore, a novel performance metric is presented, namely the *Euclidean Distance-based Accuracy (EDA)*. This new metric is capable of quantifying the amount by which incorrect predictions have failed, which could be very helpful for researchers looking to evaluate the accuracy of new classification models applied to electricity consumption time series or those with similar structures.

4. Case study

The case study comprises smart meter and weather readings from 500 households. The electricity data set comes from hourly fiscal meter readings and was pseudo anonymised to avoid data privacy issues. The data sets contain information generated for a year (2018). All the households are from a southeastern area of Spain.

27.5% of the electricity consumption corpus were *incomplete* days. To evaluate the whole procedure, the first 75% of the data corpus were selected as historical data to feed a training stage. The 25% remaining days were used as new data to be injected into a day-ahead forecasting stage. This splitting of the data corpus was done for each user. Each entry consists of:

- Electricity consumption (denoted as $y_{(d,h)}$):
  - h: hour ($0 \leq h \leq 23$).
  - d: date ($d = dd/mm/yyyy$).
- External weather conditions:
  - outdoor temperature. Obtained from a weather web service of the company Dark Sky (Anon, 0000).

5. Methodology

5.1. Overview

To evaluate the improvement introduced, this study compares the accuracy results of an hourly consumption forecasting with and without including the novel proposed behaviour feature. For this purpose, the analysis is done comparing the results of two models. The first model consists of the ANN trained to forecast hourly electricity consumption using the 24 hourly lags from the

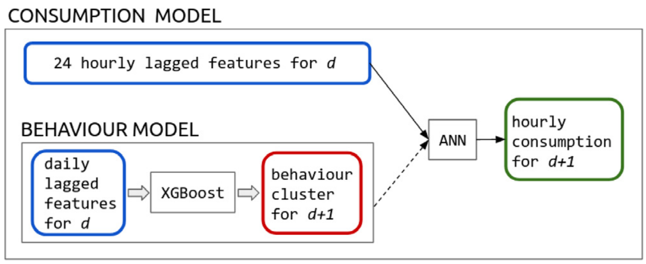


Fig. 1. Summarised structure of the forecasting model.

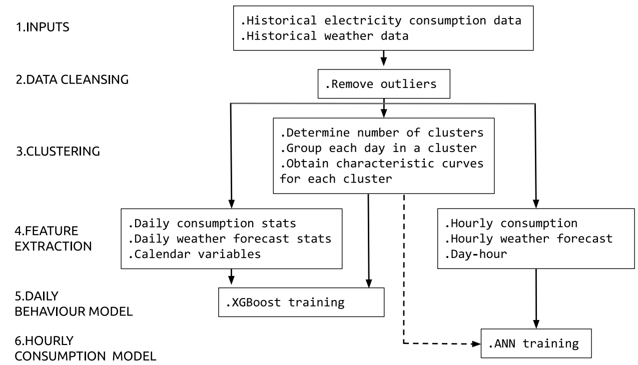


Fig. 2. Training workflow. The dotted arrow shows the features that are not considered when running only the consumption model.

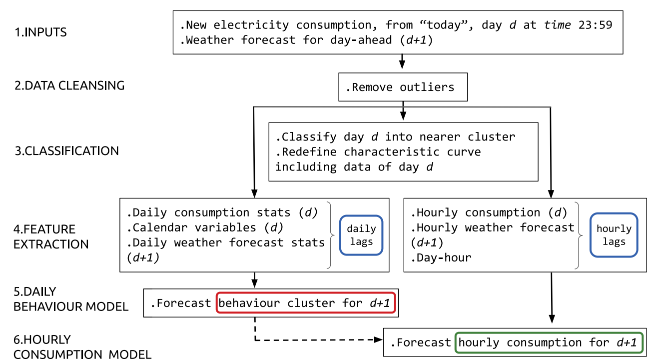


Fig. 3. Day-ahead forecasting workflow. The dotted arrow shows the features that are not considered when running only the consumption model.

previous day. And the second model consists of the consumption model to which an additional input is introduced, the prediction of the behaviour pattern. The first model is described by the scheme in Fig. 1, excluding the lower box which represents the behaviour model (linked by the dotted arrow). The second, coupled model is the complete diagram in Fig. 1.

In order to be run in an everyday real-time scenario, the proposed methodology, which consist of a workflow process, is divided into two stages: the first one is the training stage (Fig. 2) which uses historical data and; the second is the forecasting stage for the day-ahead. The last stage is shown in detail in Fig. 3. The dotted arrows in the figures show how to decouple the behaviour model from the consumption model. Depending on the computational processing availability and accuracy searched, the training stage may be executed yearly, seasonally or monthly. On the other hand, the forecasting stage is to be executed daily; this one is faster because the computational requirements are much lower. The whole process is executed independently for each customer.

5.1.1. Training

The inputs to the training stage are historical time series of electricity consumption and weather. This stage consists of data cleansing, clustering (to determine the characteristic daily electricity behaviours) and feature extraction (to identify daily variables for the behaviour forecasting model). For the case of the hourly consumption model, the feature extraction reduces to save the hourly lagged consumption, weather forecast and day-hour values; no statistics are performed. Finally, XGBoost daily behaviour model and ANN hourly consumption model are trained separately.

In the training phase there is no direct link between XGBoost and ANN. However, the clustering results are used as outputs to train the XGBoost and as inputs to train the ANN, which leads to an indirect connection. On the other hand, in the forecasting phase the connection between XGBoost and ANN is direct.

5.1.2. Day-ahead forecasting

Once the training stage has been executed for each customer, the forecasting stage is executed at the end of every day (d) to predict the next day's ($d+1$) hourly consumption. Here, new data is injected, cleaned and classified into the previously defined clusters. Classification was chosen in this stage because it is faster than clustering. Moreover, to add the new information to the process, this data is considered to redefine the characteristic curves of the clusters. Therefore, if a household has a significant change in its habitual daily consumption patterns, the model will be able to include this new pattern into the previously obtained consumption habits. Afterwards, the new daily features are extracted and introduced to the XGBoost, to predict the day-ahead behaviour cluster. Finally, this prediction is introduced, together with the hourly extracted features, to the ANN hourly consumption model, which gives an hourly consumption forecasting for the following day.

5.2. Data cleansing

When working with measurements collected from smart meter readings provided by an electric utility company, unexpected communication problems may occur, leading to periods with a lack of data or the appearance of outliers. Therefore, data cleansing is applied to the raw electricity consumption data to remove outliers. The outlier detection is performed in the following two steps:

- **Step 1**. Daily aggregate consumption must be between the limits reported by the Spanish Institute for Diversification and Energy Saving (Estudios IDAE, 2011) (2.5 kWh and 25 kWh). In addition, the maximum hourly consumption must be below 6 kWh. Days not meeting these conditions are considered outliers.
- **Step 2**. Obtaining the modified z-score (Eq. (1)) as a maximum threshold for each day over a 7-day window.

$$z_{(d,h)} = \frac{y_{(d,h)} - M(y_{(d-7,h)}, \ldots, y_{(d,h)})}{\sigma_{(d,h)}}, \tag{1}$$

where $M(y_{(d-7,h)}, \ldots, y_{(d,h)})$ is the median measured consumption over the previous 7 days at hour h and $\sigma_{(d,h)}$ is the median absolute deviation, defined in Eq. (2).

$$\sigma_{(d,h)} = M(|Y_{(i,h)} - M(y_{(d-7,h)}, \ldots, y_{(d,h)})|), \tag{2}$$

where $Y_{(i,h)} = y_{(d-7,h)}, \ldots, y_{(d,h)}$. A day d is considered an outlier if for any hour $|z_{(d,h)}| > 6$.

5.3. Clustering

The outcome of the clustering technique is a set of groups containing similar daily consumption patterns which are identified from the training data set. Clustering is applied to each user separately. Therefore, households can differ from each other in the number and shape of the daily consumption clusters.

A distribution-based clustering technique is used, in particular, the Gaussian mixture clustering is chosen. To understand how this technique works, let $A = \{\mathbf{y_1}, \ldots, \mathbf{y_d}, \ldots, \mathbf{y_n}\}$ be a sample of n independent identically distributed daily electrical consumption measurements. In this case, for example, the element $\mathbf{y_1}$ is the electrical consumption for day 1 with its hourly values in the 24 corresponding variables (each hour h is a variable). That is to say, the element $\mathbf{y_1}$ is a vector of the following type, $\mathbf{y_1} = (y_{(1,0)}, \ldots, y_{(1,h)}, \ldots, y_{(1,23)})$, where each component represents a reading of the smart meter. Assuming that the elements follow a finite mixture of a number G of Gaussian probability densities, the function f representing the distribution in the 24-dimensional space (length of variable h) is defined in Eq. (3).

$$f(\mathbf{y_d}; \boldsymbol{\Psi}) = \sum_{k=1}^{G} \pi_k N_k(\mathbf{y_d}; \mu_\mathbf{k}, \boldsymbol{\Sigma}_\mathbf{k}), \tag{3}$$

where $\boldsymbol{\Psi} = \{\pi_1, \ldots, \pi_G, \mu_\mathbf{1}, \ldots, \mu_\mathbf{G}, \boldsymbol{\Sigma}_\mathbf{1}, \ldots, \boldsymbol{\Sigma}_\mathbf{G}\}$ are the parameters of the mixture model, $N_k(\mathbf{y_d}; \mu_\mathbf{k}, \boldsymbol{\Sigma}_\mathbf{k})$ is the kth Gaussian density component with parameter vector $(\mu_\mathbf{k}, \boldsymbol{\Sigma}_\mathbf{k})$, evaluated at element $\mathbf{y_d}$, and $(\pi_1, \ldots, \pi_G)$ are the mixing weights or probabilities (with $\pi_k > 0$, $\sum_{k=1}^{G} \pi_k = 1$), and G is the number of clusters. Assuming G is fixed, the $\boldsymbol{\Psi}$ parameters must be estimated. Since the component densities are Gaussian, the clusters are ellipsoidal and centred at the mean vector $\mu_\mathbf{k}$. They have geometric features such as volume, shape and orientation, determined by the covariance matrix $\boldsymbol{\Sigma}_\mathbf{k}$.

In order to find the parameters of the mixture model ($\boldsymbol{\Psi}$) and the number of clusters (G) which best fit the model described by Eq. (3), the R package *Mclust* was used. This package provides libraries to estimate the parameters of the multi-dimensional Gaussian probability distributions which best fit the input data set. The way of achieving this is via the expectation–maximisation algorithm.

In this study, the boundaries for the number of clusters were set to $2 \leq G \leq 10$. The optimum total amount of clusters was selected by using the Integrated Completed Likelihood (ICL) criterion and the model fit was done using the Bayesian Information Criterion (BIC).

The daily characteristic curve c_h^s of each cluster s at hour h, is defined in Eq. (4). Fig. 4 shows an example of the set of characteristic curves obtained for a particular user.

$$c_h^s = \frac{1}{n_s} \sum_{d=1}^{n_s} y_{(d,h)}^s, \tag{4}$$

where n_s is the number of days belonging to cluster s and $y_{(d,h)}^s$ is the daily normalised consumption data of day d at hour $h \in \{0, \ldots, 23\}$, clustered in cluster s. When applied to this case study, the mean number of clusters obtained over all the users was 8, with standard, minimum and maximum deviation of 2, 4 and 10 respectively.

5.4. Classification

Once the characteristic electricity daily consumption clusters of each customer are identified, our aim is to classify a given day in a pre-defined cluster. Since the present research deals with large data volumes, the Euclidean distance l was selected as the classification criteria due to its simplicity. Eq. (5) shows its mathematical expression.

$$l(\mathbf{y_d}, \mathbf{c^{s'}}) = \sqrt{\sum_{h=0}^{23} \left(y_{(d,h)} - c_h^{s'}\right)^2} \tag{5}$$

$$s = \underset{s' \in \{1 \ldots G\}}{\arg\min}\, l(\mathbf{y_d}, \mathbf{c^{s'}}), \tag{6}$$

where $\mathbf{y_d}$ is the vector containing the 24 consumption readings for the day d and $\mathbf{c^{s'}}$ is the vector containing the 24 obtained values for the characteristic curve of each cluster s'. The new day is classified into the cluster s that minimises the Euclidean distance l.

Classification was chosen to replace clustering in the forecasting stage because it is faster. However, to avoid losing the new information added every day to the process, such as observed in similar methodologies (Yildiz et al., 2018), this data is considered to redefine the characteristic curves of the clusters. This every day redefinition of the characteristic patterns allows the model to cope with changes in typical consumption habits. In this way, the model will take only some days to adapt to the new pattern, which will depend on how big the difference in consumption is, but it will gradually reshape the clusters without the need of clustering again. Many causes can generate changes in consumption patterns (these could be: Covid, having a new baby, retirement, etc.), so it is important that the methodology is prepared to deal with this aspect.

5.5. Feature extraction

This step consists of selecting and modifying the time series to give rise to some characteristic features that will later serve as input variables for the prediction model. Feature extraction reduces dimensionality and highlights aspects of the time series, allowing a better understanding of the dependence of user behaviour with different features obtained from the time series.

Daylight imposes a natural rhythm on human behaviour, therefore, to extract daily related features is an obvious choice. Apart from the consumption and weather data which is extracted from the data corpus, calendar features were added; including calendar information has proven to enhance electricity forecasting models (Ramos et al., 2020). Therefore, every day is characterised by:

- **consumption daily statistics**. Made up of quantiles 10 and 90, standard deviation, mean daily consumption, mean consumption during the night (00:00–05:00), mean consumption during the morning (06:00–12:00), mean consumption during the afternoon (13:00–19:00) and mean consumption during the evening (20:00–23:00).
- **weather daily statistics**. Made up of: mean temperature, maximum temperature and minimum temperature.
- **calendar**. Made up of: weekday or holiday; summer season, winter season or midseason (merges autumn and spring).

Once the features are extracted, they are transformed using base splines as a strategy to make them dimensionless. Creating dimensionless features is an important task when the input variables to the model represent different physical magnitudes that range between significantly different limits. Each identified feature is transformed using base splines $B(x)$ of 3 knots that satisfy the recursive expression of Cox–de Boor (Eq. (7)).

$$\begin{aligned} B_{i,0}(x) &= \begin{cases} 1 & \text{for } t_i \leq x < t_{i+1} \\ 0 & \text{otherwise} \end{cases} \\ B_{i,p}(x) &= \tfrac{x-t_i}{t_{i+p}-t_i} B_{i,p-1}(x) + \tfrac{t_{i+p+1}-x}{t_{i+p+1}-t_{i+1}} B_{i+1,p-1}(x) \end{aligned} \tag{7}$$

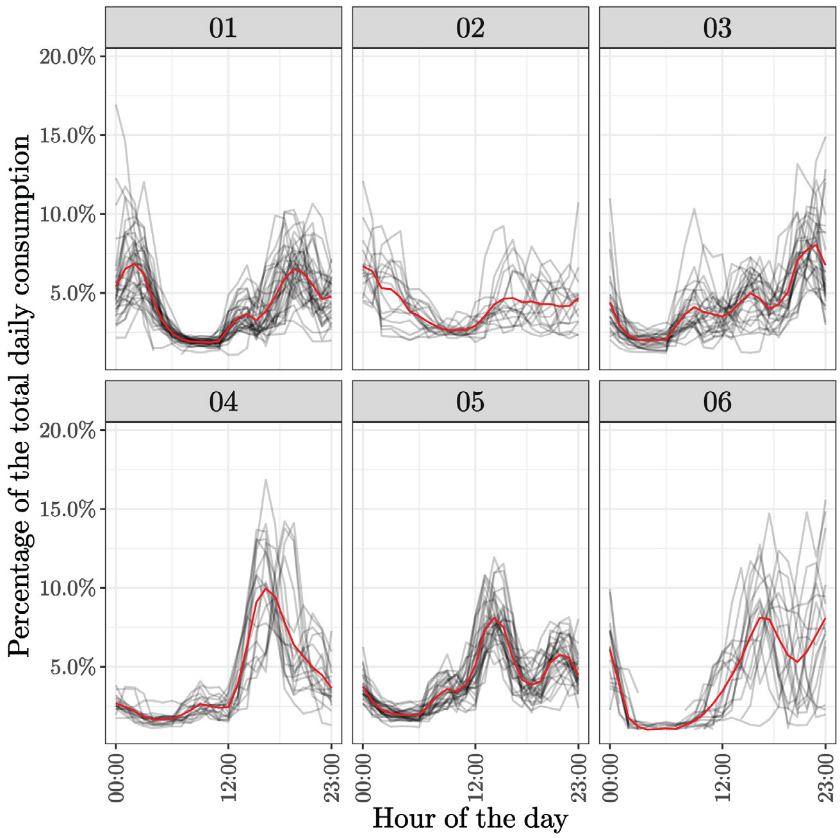


Fig. 4. Characteristic curves for each of the 6 clusters obtained (in red). Normalised curves (in black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where $i = \{1, 2, 3\}$ are the knots, p the polynomial order and t_i the specified boundary and interior knots. The boundary knots are chosen as the minimum and maximum values of the feature, and the interior knot corresponds to its mode. This way of creating dimensionless inputs works better than simply normalising each feature because, in this case (by choosing the mode as the interior knot), the most recurrent values are put together.

Base splines of order $p = 1$ were chosen. In this way, piecewise linear polynomials are obtained. They take values within the interval [0, 1] in the range defined by the knots, and zero values out of this range. The transformation of the variables into base splines is a way of stratifying the range of values. Once all the variables have been transformed into base splines, the characterisation becomes dimensionless.

Fig. 5 shows an example for the feature *night*. In Fig. 5(a), the time series of the variable is plotted, Fig. 5(b) shows its histogram and Fig. 5(c) shows the B-splines. In all the graphs, the mode, which is used as the interior knot for the B-spline calculation, is represented by a dotted line. B-spline1 represents the values below the mode, B-spline2 the values near the mode, and B-spline3 the values above the mode.

Finally, all the features are lagged so that each day entry contains the information from the previous ones. Lags are defined as the values the features and their base spline transformation took the previous 1, 2, 3 and 7 days.

5.6. Daily behaviour model

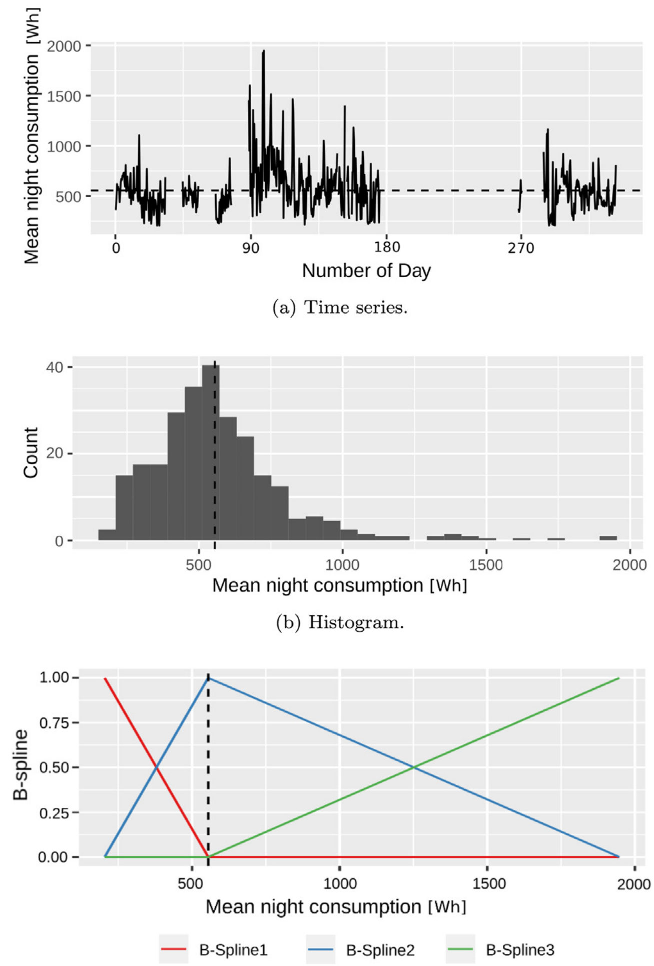
In this step, we are interested in forecasting the day-ahead behavioural cluster for each user. Depending on the data available regarding the previous days, two day-states are defined: complete and incomplete. In a training stage, the behavioural prediction model is trained only with complete days. The XGBoost package is used since its implementation supports automatic handling of missing data (Chen and Guestrin, 2016). Considering the high amount of *incomplete* days (27.5%), this aspect was of great importance when choosing among different algorithms.

XGBoost is an efficient implementation of gradient boosting, typically used with decision trees. The boosting algorithm is an ensemble learning method that combines weak classifiers (such as the decision trees chosen in this study) to achieve a strong final classifier. This is done by iteratively training weak classifiers and adding them. When they are added, they are weighted in a way that is related to the weak learners' accuracy. Each decision tree learns from the previous one and affects the next tree to improve the model (Friedman, 2001). Gradient boosting models have been compared with other machine learning algorithms (such as K-Nearest Neighbour and Support Vector Machines), and it has been found that the boosting technique outperforms these in terms of accuracy and robustness (Bennett et al., 2007; Yu et al., 2010). Sepulveda et al. (Sepulveda et al., 2021) show a methodology in which gradient boosting is used to predict energy consumption in a massive scenario.

Eq. (8) describes a tree ensemble model which predicts the dependent variable $\hat{s_d}$ using the independent variables x_d and J additive functions.

$$\hat{s_d} = \sum_{j=1}^{J} f_j(\mathbf{x_d}) \tag{8}$$

In this research, $\hat{s_d}$ is the predicted behaviour cluster for day d, f_j represents an independent tree structure with leaf scores in the space of trees, and $\mathbf{x_d}$ is the vector of independent variables, which, in this study, is composed of the lagged features



(a) Time series.

(b) Histogram.

(c) Decomposition in the three B-splines.

Fig. 5. Base spline transformation for the variable *night*. The mode is represented with a dotted line.

for day d described in Section 5.5 (consumption daily statistics, weather daily statistics and calendar, with their respective B-spline transformations).

XGBoost performs the parameter estimation by minimising the following loss function:

$$\mathcal{L}(\Phi) = h(\hat{s_d}, s_d) + \sum_{j=1}^{J} \Omega(f_j), \tag{9}$$

where Φ are the model parameters that have to be estimated, Ω is a term for penalising the complexity of the model and h is the following log loss function:

$$h(\hat{s_d}, s_d) = -\frac{1}{N_d} \sum_{d=1}^{N_d} \sum_{s=1}^{G} u_{ds} \log p_{ds}, \tag{10}$$

where N_d is the number of days to train the model, G is the number of possible clusters for that user, u_{ds} is a binary indicator of whether or not cluster s is the correct classification for day d, and p_{ds} is the model probability of assigning cluster s to day d. The log loss function quantifies the accuracy of a classifier by penalising false positives.

The tuning of XGBoost parameters was performed for each user using the *mlr* R package (Bischl et al., 2016). This strategy was chosen because the implementation aims to be used in a massive realistic scenario. Therefore, parameter tuning needed to be automated.

Table 1
ANN characteristics.

Parameter	Default value
Number of hidden layers	1
Number of neurons in hidden layer	100

5.7. Hourly, day-ahead electricity consumption forecasting model

An ANN was chosen to forecast the day-ahead hourly electricity consumption. Multi-layer perceptron (MLP) was used as the ANN approach. A separate ANN model was trained for each household. However, since for all of the consumption forecasting model the size of the input layer and the size of the output layer is the same, it was considered redundant to run the tuning of the number of neurons for each of the users. To obtain the optimal number of neurons, 5-fold cross-validation was ran for 10 different users, obtaining a number near 100 for all of them. In the hidden layer, the activation function used was the logistic function, $f(x) = 1/(1 - e^{-x})$. Table 1 contains the values assigned to the ANN hyper-parameters. The weights were obtained using backpropagation. The Scikit-learn python machine learning library was used to implement the ANN (Pedregosa et al., 2011).

5.8. Performance metrics

There are several metrics to evaluate classification algorithms (Tharwat, 2018). Although *accuracy* is the most widely-used performance metric, it was considered incomplete for this research. Since the elements being classified are daily consumption curves comprised of time series with 24 values, such binary metric provides minimal understanding about the degree of deviation regarding the right characteristic consumption pattern. That is why the need arose to search for a new metric. This new metric should not only detect correct/incorrect predictions but should also quantify them. Therefore, a new performance metric is presented, namely the *Euclidean Distance-based Accuracy (EDA)*. *EDA* quantifies the proximity of the electricity daily consumption curves. Using the Euclidean distance (Eq. (5)), the daily metric *EDA*, corresponding to cluster s and predicted as cluster $\hat{s}$, is defined in Eq. (11).

$$EDA = \frac{l(\mathbf{y_d^s}, \mathbf{c^{\hat{s}}}) - l(\mathbf{y_d^s}, \mathbf{c^s})}{l(\mathbf{y_d^s}, \mathbf{c^s})} \tag{11}$$

The numerator is equal to the difference between the distance from a daily consumption curve ($\mathbf{y_d^s}$) to the characteristic curve of the predicted cluster $\hat{s}$ ($\mathbf{c^{\hat{s}}}$), and the distance from the same consumption curve ($\mathbf{y_d^s}$) to the characteristic curve of the cluster s being classified ($\mathbf{c^s}$). This difference is divided by the distance between $\mathbf{y_d^s}$ and $\mathbf{c^s}$. Accordingly, *EDA* represents how far the predicted characteristic curve is from the classified characteristic curve in terms of the second one.

A reliable metric should also be identified to evaluate the improvement introduced in the consumption model when including the predicted behaviour cluster. The most commonly used metrics in the literature are the Mean Absolute Percentage Error (MAPE) and the Normalised Root Mean Square Error (NRMSE). The expression of the daily MAPE_d is defined in Eq. (12) and the daily NRMSE_d in Eq. (13). For an ideal model, $\text{MAPE}_d = \text{NRMSE}_d = 0$.

$$\text{MAPE}_d = 100 \times \frac{1}{n} \sum_{h=0}^{23} \left| \frac{\epsilon_{(d,h)}}{y_{(d,h)}} \right| \%, \tag{12}$$

$$\mathrm{NRMSE}_d = 100 \times \frac{1}{\bar{y}_{(d,h)}} \sqrt{\frac{\sum_{h=0}^{23} (\epsilon_{(d,h)})^2}{n}} \%, \tag{13}$$

where $n = 24$, $\epsilon_{(d,h)} = \hat{y}_{(d,h)} - y_{(d,h)}$, $\hat{y}_{(d,h)}$ is the forecasted consumption and $\bar{y}_{(d,h)}$ is the mean daily consumption.

5.9. Feature explanation

Model interpretability is very useful for electricity network operators, demand response aggregators and electricity-trading companies because, in this case, it would allow them a better understanding of each residential user's behaviour. Therefore, the SHapley Additive exPlanations (SHAP) method was used (Lundberg and Lee, 2017) to analyse the contribution of each individual feature to the model.

SHAP is an approach based on game theory to explain the output of any data-driven model. It connects optimal credit allocation with local explanations using the classic Shapley values from game theory. SHAP is used to understand how a single feature affects the output of the model. It shows improved computational performance and better consistency than previous approaches. In SHAP, the importance of feature i, I_i (for $i = 1, \ldots, n$), is calculated according to Eq. (14).

$$I_i = \sum_{j=1}^{n} \left| \phi_i^{(j)} \right|, \tag{14}$$

where $\phi_i^{(j)}$ is the marginal SHAP contribution of feature i compared to feature j.

6. Results

6.1. Validation of the new metric EDA

As mentioned previously, *EDA* is used to evaluate the accuracy of the day-ahead behavioural cluster model. This section shows an analysis using *EDA* to show that this parameter fits the requirements stated in Section 5.8. Six days were selected out of one user (which is different from the one studied in Fig. 4) to elaborate the validation analysis of *EDA*. These days were chosen because they show different scenarios which allow the complete understanding of how the new metric works. Figs. 6(a) to 6(f) show the real consumption curve (black), the classified curve (green), the predicted curve (red) and the curves of the remaining clusters (grey). The clusters were generated after running the Gaussian mixture model clustering and calculating the daily characteristic curves according to Eq. (4).

In the first two (Figs. 6(a) and 6(b)), the predicted cluster is equal to the classified one, meaning forecasting success ($s = \hat{s}$; $EDA = 0$). In Fig. 6(a), the predicted and classified curves are equal and are both similar to the real consumption. On the other hand, note that in Fig. 6(b), these curves are not as near to the real consumption curve as in the previous case, but still, the predicted cluster is the nearest to the real consumption curve of the whole set of clusters.

Small (but non-zero) *EDA* suggests that even though the prediction model could have performed better, the actual result is not "so far" from the expected one in terms of the distance between the classified and the real consumption curve. This occurs when the distance from the real curve to the classified curve is similar to the distance from the real curve to the predicted one. If the classified cluster is sufficiently near to the real curve, then a small *EDA* will imply that the predicted curve is near enough to the real curve. This case is illustrated in Fig. 6(c). On the other hand, if the classified cluster is far from the real curve, a small *EDA* means that it is impossible to forecast a cluster with a characteristic curve close to the daily consumption. This is basically because there is no such characteristic curve among the set generated by the clustering algorithm, but it is still performing "well". The latter may occur when the user generates a new consumption pattern that is inserted into the day-ahead forecasting stage, and it cannot be represented by any of the clusters defined over the historical data in the training stage. Fig. 6(d) shows an example of this case. All in all, small *EDAs* mean good performance of the model.

Finally, *EDA* is large when the predicted cluster is far from the classified one in terms of the distance between the classified and real curves (see Figs. 6(e) and 6(f)). After analysing several cases, the forecasting success/failure frontier was set as $EDA = 1$.

Taking everything into account, three groups of days were defined:

- **perfectly assigned** ($EDA = 0$): 33.2% of the days,
- **correctly assigned** ($0 < EDA \leq 1$): 46.5% of days,
- **incorrectly assigned** ($1 < EDA$): 20.3% of days.

In conclusion, *EDA* allows a broader study, which understands that forecasting success is not restricted solely to the fact that the predicted curve is exactly the same as the classified curve.

6.2. Model accuracy evaluation

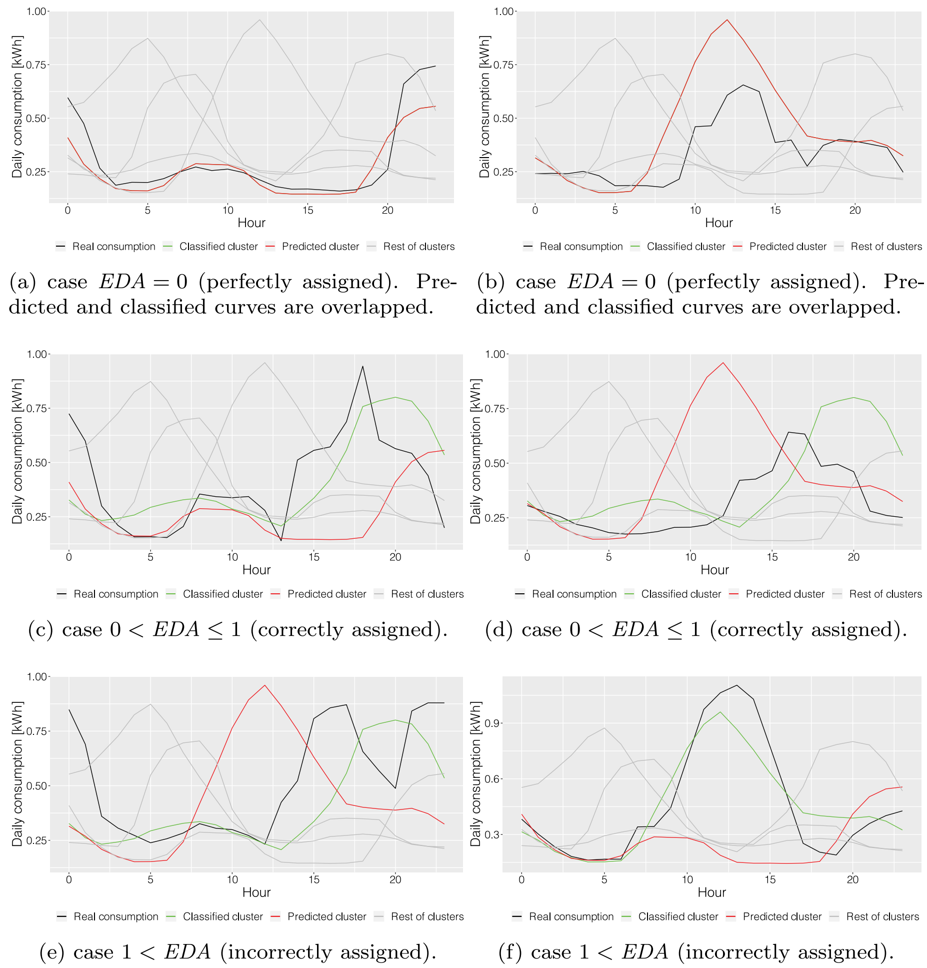
Table 2
Comparison of performance metrics.

Model	MAPE_d (%)	NRMSE_d (%)
Consumption model	63.4	56.4
Consumption model + Behaviour model	**56.2**	**47.0**

Table 2 presents a comparison between the ANN model and the same ANN model when including the behavioural cluster forecasted as one of the inputs, as described in Section 5.1. The accuracy evaluation was performed using MAPE_d and NRMSE_d, the best performance values are highlighted in bold. As observed in Table 2, the MAPE_d and NRMSE_d were reduced (7% and 9% respectively) when including the behaviour model. On the other hand, the average *EDA* obtained was 0.633, which means a good outcome since it is within the correctly assigned rank.

The performance metrics were further analysed for each of the group of days previously defined according to the *EDA* obtained (perfectly, correctly and incorrectly assigned). Tables 3–5 present the MAPE_d and NRMSE_d obtained for each group. Table 3 shows that when the forecasting of the behaviour model is exact ($EDA = 0$), the metrics improved significantly. The averaged MAPE_d was reduced by 20% and the NRMSE_d was reduced by 17%. Table 4 shows that for the correctly assigned group, the differences in MAPE_d and NRMSE_d are smaller (5% and 8%, respectively), but the outcomes improved with the behaviour model. Finally, Table 5 shows the analysis for the incorrectly assigned group, with little differences in MAPE_d between both models (5%) and undetectable differences for NRMSE_d. Overall, it is clear that the performance metrics improved when the behaviour model was included, with the perfectly assigned group being the one that evidenced the most remarkable results.

When comparing with performance metrics obtained for aggregated forecasting models (such as when applied to the consumption of a whole city), these MAPE_d and NRMSE_d values might seem high. However, when contrasting with the metrics for forecasting models of individual residential households, these metric values fall within valid ranges. In Yildiz et al. (2018), for example, it is observed that the obtained mean NRMSE_d is around



(a) case $EDA = 0$ (perfectly assigned). Predicted and classified curves are overlapped.

(b) case $EDA = 0$ (perfectly assigned). Predicted and classified curves are overlapped.

(c) case $0 < EDA \leq 1$ (correctly assigned).

(d) case $0 < EDA \leq 1$ (correctly assigned).

(e) case $1 < EDA$ (incorrectly assigned).

(f) case $1 < EDA$ (incorrectly assigned).

Fig. 6. Daily consumption curves for one user.

Table 3
Perfectly assigned (*EDA* = 0).

Model (for **perfectly assigned**)	$MAPE_d$ (%)	$NRMSE_d$ (%)
Consumption model	66.0	54.8
Consumption model + Behaviour model	**46.1**	**38.1**

Table 4
Correctly assigned (average EDA = 0.439).

Model (for **correctly assigned**)	$MAPE_d$ (%)	$NRMSE_d$ (%)
Consumption model	61.5	55.9
Consumption model + Behaviour model	**56.0**	**48.2**

60%, with particular values ranging between 40% and 90%. In our case, the obtained $NRMSE_d$ for the perfectly assigned group equals 38% which is slightly lower than the best results of the research performed by Yildiz et al. (2018).

Fig. 7 shows two example days which were chosen to help in the interpretation of the performance metrics analysed previously. Fig. 7(a) represents a day for which both the Consumption model and the Consumption + Behaviour model obtained the worst performance metrics ($MAPE_d \sim 55\%$ and $NRMSE_d \sim 60\%$). And Fig. 7(b) represents a day for which both the Consumption model and the Consumption + Behaviour model obtained the best performance metrics ($MAPE_d \sim 37\%$ and $NRMSE_d \sim 35\%$).

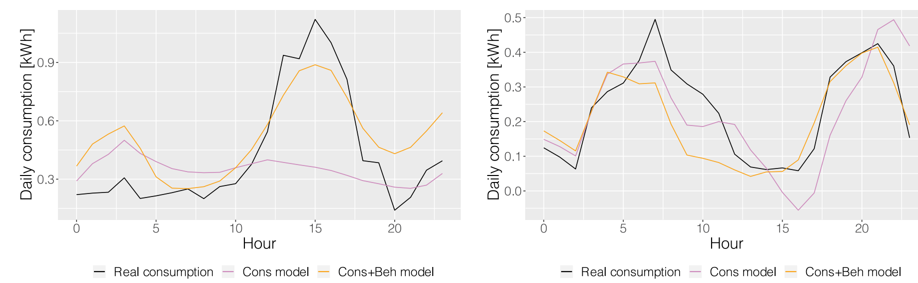
Table 5
Incorrectly assigned (average EDA = 2.11).

Model (for **incorrectly assigned**)	$MAPE_d$ (%)	$NRMSE_d$ (%)
Consumption model	78.0	59.9
Consumption model + Behaviour model	**63.8**	**59.1**

6.3. Feature explanation

For this analysis, 217 users, which obtained the 75th percentile of *EDA* lower or equal than 1, were selected. The following figures sort features and lags by their mean absolute SHAP value (obtained with Eq. (14)), therefore getting an overview of which are most important for the model. The feature importance ranges from 0 to 1.

Fig. 8 shows a bar plot of the feature importance in this data set for the daily behaviour model. As can be appreciated, the calendar variables (*week* and *season*) were the ones with greatest importance. This matches the intuitive expected user's behaviour, which depends mainly on the day of the week and the season of the year. It can also be seen that maximum and minimum temperatures were more important than the mean daily temperature. This could be due to the fact that the mean daily temperature is highly correlated with the season, while maximum and minimum temperatures have higher daily variation within each season. The importance of external temperature revealed in this analysis reinforces the significance of including the weather forecast data set which is sometimes not used in similar approaches (such as in Bian et al. (2020)).



(a) Example day with worst metrics. (b) Example day with best metrics.

Fig. 7. Daily consumption curves for one user.

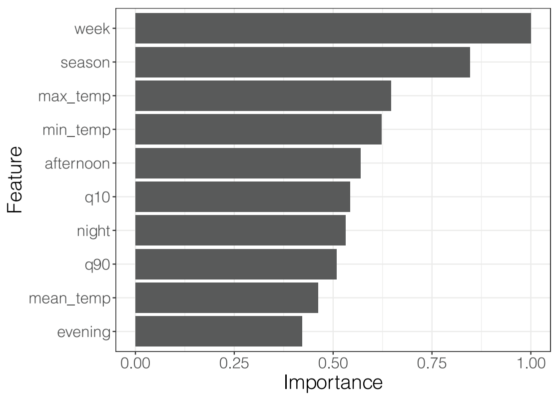


Fig. 8. Aggregate feature importance for $EDA \leq 1$.

Among the consumption daily statistics, the *afternoon* (13:00–19:00) mean consumption stands out as the most relevant. This could be related to the fact that, according to the normal Spanish working time schedules, most people arrive home during the afternoon, making the consumption behaviour more evident during these hours. During the remaining hours, the consumption is formed mainly of the so-called standby mode, which consists of the base consumption of all the domestic appliances which are on while the household is unoccupied or independent of human behaviour.

Following the *afternoon*, the *quantile 10* consumption has the highest impact on the model. Finally, *night* (00:00-05:00), *quantile 90* and *evening* consumption are the least important features. The remaining variables are not shown due to their low importance.

Fig. 9 shows the SHAP plot for the daily lags. Interestingly, *lag 1* is the less predominant one, showing that the users' behaviour does not depend much on the previous day's information. The ordered sequence found was: *lag 3*, *lag 2*, *lag 7* and *lag 1*. This shows a particular component of behaviour memory in which their past behaviour influences the customers' conduct. This conduct cannot be modelled by a Markov chain because it does not follow an ordered sequence. This reinforces the use of daily-frequency features as inputs. Higher granularity (i.e. hourly), would only have added redundant information.

For this study the feature importance analysis has been done aggregating the models for all the users, because a thorough analysis including details for each user would have covered too much space. But in practice, this same feature importance analysis could be applied to each user separately, providing energy utilities a detailed understanding of each household.

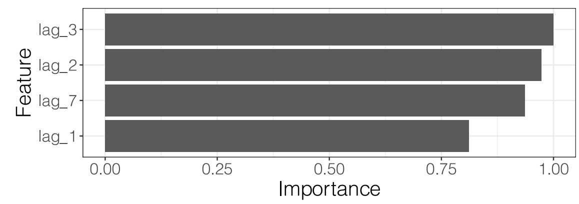


Fig. 9. Aggregated feature importance of the models obtained for 217 selected users which obtained $EDA \leq 1$.

7. Discussion

A transition towards distributed RES is taking place, and tools to accurately forecast the individual electricity consumption are essential to guarantee the proper synchronisation of the grid. However, unlike aggregate consumption, which has been exhaustively discussed in previous research, studying the residential sector in an individual way is a tough task because user behaviour plays a major role. The existing studies focusing on individual electricity consumption forecasting models include the effect of user behaviour. Still, they do it by using extra information from surveys or sub-metering, which are impossible to set massively. Moreover, most of the studies lack the basis to make the implementations work on a large scale when the forecasting results need to be obtained within a reasonably low computational time.

This work presented a methodology focused on deeper assessment of the characteristic stochastic behaviour of residential household customers. The methodology operated very well over a large number of residential households with hourly sampling frequency data and a high percentage of *incomplete days* at an individual level. In addition, when comparing the electricity consumption forecasting models, including the behaviour component has improved the performance metrics ($MAPE_d$ and $NRMSE_d$), opening the door to more effective real-time services. Regarding applicability, when running the forecasting for the following day over the complete set of 500 residential households (in R environment using a 16 threads processor - AMD2700x), the processing time was 13 minutes. This shows that the novel methodology runs in an acceptable time to be used in real applications.

Moreover, the novel metric presented (*EDA*) allowed a broader study by understanding that forecasting success is not restricted solely to the fact that the predicted curve is exactly the same as the classified curve. Thanks to *EDA*, cases in which the model has not succeeded are clearly distinguished. Doing special insight of forecasting failure days, such as those shown in Figs. 6(e) and 6(f), it is observed that the model predicts a very different cluster (even though the real pattern has a similar shape to a cluster from the set). This failure might owe to changes in user's scheduling (for example, a person who decides to change the day in which she/he leaves the house to perform hobby activities). It is evident that the model will not be able to predict unexpected scheduling changes. This is the greatest limitation of the presented model. Once again, the reason is that human behaviour has an inherent stochastic nature which makes it highly unpredictable. A possible solution to this problem would be including human scheduling through internet communication, which would introduce an alert in the model when changes are expected to occur. However, implementing this type of communication framework in a massive scenario would add considerable complexity to the methodology and would require total commitment from users.

Direct comparisons should not be taken as strict benchmarks due to possible differences in electricity consumption data sets, climate conditions, etc. However, they can serve as one more point of comparison. In this sense, when comparing with similar studies in residential households (Yildiz et al., 2018), it is observed that the mean $NRMSE_d$ is around 60%, showing that $NRMSE_d = 38\%$ for the perfectly assigned group represents a significant improvement. This reinforces the good performance of the presented methodology. Another interesting result is that $MAPE_d$ and $NRMSE_d$ increased with *EDA* in the Consumption + Behaviour model. This shows that the model presented improves when the user behaviour is correctly predicted. Which reinforces the importance of taking into account the behaviour aspect.

Finally, analysing Fig. 7(a), it is evident that both curves fail in predicting the magnitude of the real consumption's peaks. The Consumption + Behaviour model overestimates the first daily peak and underestimates the second. On the other hand, the Consumption model is not even able to predict that two daily peaks will occur. Fig. 7(b) shows that even though both models could predict the time of the day when the significant peaks occurred, the Consumption + Behaviour model replicated best. On the whole, from observing both figures, it can be appreciated that the overall daily trend (meaning the moment when the peaks and the valleys occur) is better reproduced by the Consumption + Behaviour model.

To conclude, this methodology sets the basis for massive user-centred analyses, which may be very useful for electricity network operators, demand-response aggregators or electricity-trading companies. Providing them with better disaggregated predicting tools will allow the development of individualised control strategies. This, added to the active participation of residential consumers, will lead to significant energy savings.

8. Conclusions and future work

This research contributes to filling the individual electricity forecasting knowledge gap by combining several data-driven techniques. The strategy is divided into two stages. The first one is a training stage where historical data is used to identify representative clusters, applying Gaussian mixture models, and the training of an XGBoost model to predict the day-ahead cluster. This is followed by a day-ahead forecasting stage in which the clustering is replaced by classification to reduce the computational cost and time. The final result is the online prediction, provided by the XGBoost model, of a behaviour component for each user. This outcome is finally used as one of the inputs of an ANN day-ahead electricity consumption forecasting model.

A novel performance metric to evaluate classification algorithms applied to time series, named *EDA*, was defined and put into practice. *EDA* resembles the classification *accuracy*, but it is a real number that enables a more thorough evaluation. The boundary between the forecasting success/failure was set to $EDA = 1$. The average *EDA* obtained in the case study was 0.633, which is considered a good outcome since it is below 1.

An evaluation of the accuracy improvement generated over a day-ahead electricity consumption forecasting model is reported. The comparison was performed between an XGBoost plus ANN, and one considering only an ANN. The results showed that the averaged $MAPE_d$ and $NRMSE_d$ were reduced by 7% and 9% respectively. Moreover, for the perfect days ($EDA = 0$), the averaged $MAPE_d$ was reduced by 20% and the $NRMSE_d$ was reduced by 17%.

In addition, the behaviour model allows for an importance analysis using SHAP. Therefore, this analysis was carried out for the models corresponding to selected users. The calendar variables (*week* and *season*) showed to be the most important ones, followed by the daily maximum and minimum temperatures and the *afternoon* (13:00–19:00) mean consumption. Regarding the day lags, the decreasing order of importance was: *lag 3*, *lag 2*, *lag 7* and *lag 1*. This shows a non-ordered component of behaviour memory which is better modelled by daily-features extraction than by Markov chain models. This kind of analysis could be very useful for electricity trader companies, because it would allow them a better understanding of each customer.

The presented methodology is promising for practical applications, and electricity companies could use this tool for a detailed understanding of each separate household to provide additional and ad-hoc services. Furthermore, a future trend is to develop R packages implementing the novel methods, thus providing a means to spread this research across the industrial and private electricity community.

CRediT authorship contribution statement

Florencia Lazzari: Conception and design of study, Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing. **Gerard Mor:** Conception and design of study, Analysis and/or interpretation of data, Writing – original draft. **Jordi Cipriano:** Acquisition of data, Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing. **Eloi Gabaldon:** Acquisition of data. **Benedetto Grillone:** Acquisition of data. **Daniel Chemisana:** Acquisition of data. **Francesc Solsona:** Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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